

A Parameter-Free Honest-Stance Attestation for Conventional Superconductors:

The Niobium BCS Universal-Gap-Ratio as a Cross-Domain `absorbed=true` Gate

demiurge / hexa-lang substrate authors*

2026-05-21

Abstract

We establish the first `absorbed=true` attestation for a superconducting material in the `demiurge/RTSC.md` 4-tier expansion framework. The attestation rests on the BCS *universal gap-to-temperature ratio* $2\Delta(0)/(k_B T_c) = 2\pi e^{-\gamma_E} \approx 3.5276$, a parameter-free first-principles prediction of weak-coupling Bardeen-Cooper-Schrieffer theory. Comparing this prediction against an inverse-variance-weighted consensus of five independent tunneling-spectroscopy measurements of pure niobium yields a relative error of 0.309%, well below a pre-registered 5% threshold. The resulting record passes `GATE_CLOSED_MEASURED` and emits `absorbed=true`. The same model-vs-measured-oracle pattern that yielded the project's first `absorbed=true` (a solar-pyranometer GHI parity) thus generalises across domains. We argue that the `RTSC.md` §8.8 invariant which forbids `absorbed=true` for room-temperature claims (LK-99, n=6 closed-form, etc.) remains intact: niobium is unambiguously LTS, and the attestation does *not* validate any room-temperature hypothesis.

1 Introduction

The empirical landscape of superconductivity is partitioned by an asymmetry of confidence. On one side stands a century of replicated, multi-laboratory measurements on conventional metals (Hg, Pb, Nb), low- T_c alloys (NbTi, Nb₃Sn) and high- T_c cuprates (YBa₂Cu₃O₇, BSCCO) whose phenomenology is so reproducible that university teaching laboratories rediscover it every semester. On the other side are room-temperature claims — most recently LK-99 [6, 7] — which after the first announcement of a 300 K resistive transition fail to replicate [9], leaving a long tail of papers, retractions, and contested measurements.

A framework that grants the second class the same epistemic status as the first is a framework that has lost the ability to discriminate. The `demiurge` substrate, codified in `RTSC.md` §8.7–§8.8, encodes this asymmetry as a code-level invariant: a machine-readable `absorbed` flag that defaults to `false` and is granted `absorbed=true` only when a strict, multi-criterion gate (`GATE_CLOSED_MEASURED`) is satisfied. Crucially, this gate is enforced by type system and unit test — not merely by reviewer goodwill.

Until the work reported here, the substrate hosted exactly one `absorbed=true` record, in the energy domain: a solar pyranometer oracle (NREL MIDC BMS, Golden CO, 2024-06-15) compared against the `pvlib` Ineichen clearsky model, with `mean_rel_err` = 4.99% under a pre-registered 5% threshold [10, 13].

*Corresponding repo: <https://github.com/dancinlab/demiurge>, <https://github.com/dancinlab/hexa-lang>.

This paper extends `absorbed=true` into the superconductor domain (`rtsc.md`) by exploiting a single, deep observation: the BCS universal gap-ratio is parameter-free. The model side knows nothing about niobium specifically; it is a constant derived from a contour integral in weak-coupling theory. The measurement side is canonical tunnelling spectroscopy on niobium across five independent groups spanning 1960–1996. The parity passes the same 5% threshold to 0.309%, an order of magnitude inside the gate.

Contributions.

1. We introduce the `nb_bcs_absorbed_attestation_producer` (§4) which emits a typed JSON record matching the pre-existing pyranometer schema bit-for-bit (§6).
2. We document a 5-paper inverse-variance consensus on $2\Delta(0)/k_B T_c$ for niobium and verify the BCS prediction to 0.309% (§5).
3. We discuss the cross-domain generality of the *measurement-oracle parity* pattern — solar GHI and BCS gap-ratio differ in physics but share schema — as a portable template for future `absorbed=true` gating (§6).
4. We re-affirm that the `RTSC.md` §8.8 invariant forbidding `absorbed=true` for room-temperature SC *hypotheses* remains untouched (§3.2); we explicitly enumerate the four-source honest-negative for LK-99 (§7).

2 Background: the BCS universal gap-ratio

In weak-coupling BCS theory [2], the zero-temperature energy gap $\Delta(0)$ and the critical temperature T_c of a clean, phonon-mediated, isotropic *s*-wave superconductor obey

$$\frac{2\Delta(0)}{k_B T_c} = 2\pi e^{-\gamma_E} \approx 3.5276, \quad (1)$$

where $\gamma_E = 0.5772\dots$ is the Euler–Mascheroni constant.

The derivation [16, Ch. 3] solves the coupled self-consistent equations for the gap and the chemical potential within the BCS effective Hamiltonian, and the constant on the RHS of Eq. (1) emerges with *no* material-specific input. The microscopic details of phonon spectrum, electron-phonon coupling λ , Coulomb pseudo-potential μ^* , and Fermi surface anisotropy all factor out of the ratio — they reappear only when one tries to compute T_c and $\Delta(0)$ *individually*.

For our purposes this means: the universal ratio is a *blind test* of BCS theory. Any agreement (or disagreement) between prediction and measurement is informative without circularity [12, Ch. 4 §3].

Range of validity. The ratio is renormalised upward for strong-coupling materials (Pb ≈ 4.3 ; Nb₃Sn ≈ 4.1 ; MgB₂ small gap ≈ 1.7 and large gap ≈ 4.5 [15]). It does not apply to unconventional *d*-wave cuprates, where $2\Delta(0)/k_B T_c \approx 4$ –8 depending on doping [3]. Niobium is the cleanest weak-coupling textbook test case; $\lambda \approx 0.82$, and the strong-coupling correction to the ratio is below the typical measurement scatter.

3 The honest-stance framework

3.1 Four-tier expansion path

`RTSC.md` §8.7 specifies four independent layers that the `demiurge` substrate must satisfy before a material claim can even be considered for absorption:

- Tier 1 – Computational synthesis.** First-principles predictions of T_c , H_{c2} , gap functions. In the present substrate this is implemented as `sim.adapter.py` (Python) and `sim.hexa` (hexa-native), the latter agreeing with the former to `libm` precision ($\Delta T_c = 0.0000\text{ K}$ across BCS, McMillan, Allen-Dynes [1]). Tier 1 alone never grants `absorbed=true`: predictions are not measurements.
- Tier 2 – Synthesis-recipe-as-record.** Typed JSON recipes citing reagents, conditions, and `replicated_by_ independent_labs`. Industry-mature recipes (REBCO MOCVD, NbTi PIT wire) carry replication counts ≥ 5 ; LK-99 carries 0.
- Tier 3 – Measurement ingest.** External instrument data as typed records: $R(T)$, Meissner $\chi(T, B)$, AC susceptibility, specific heat, $J_c(B, T, \theta)$, $H_{c2}(T)$. Each carries a `replication_count` ≥ 1 ; independent-lab cross-checks raise it.
- Tier 4 – Falsifier dispatch.** A typed verdict that consumes a (T1, T2, T3) triple and emits a per-falsifier PASS / FAIL / SKIPPED-MISSING-INPUT decision. Six falsifiers, three from RTSC and three from conventional SC, map to the 9-test characterisation of [11]. Tier 4 PASS alone does *not* flip `absorbed=true`; an attestation gate (this paper) must agree.

3.2 The `absorbed=false` invariant

RTSC.md §8.8 encodes a single hard constraint: `absorbed=true` is forbidden for any room-temperature SC hypothesis that has not been independently replicated. This is not a heuristic or stylistic preference — it is a code-level invariant. The Tier 4 dispatcher `MaterialFalsifierDispatch` (Swift) carries a unit test `testAbsorbedAlwaysFalseEvenWithReplication` that asserts: *however many replications, however many PASSing falsifiers, the dispatcher’s output record must have `absorbed=false`.*

The only path to `absorbed=true` is therefore a *separate* producer that emits, on its own authority, a record whose `measurement_ gate` is `GATE_CLOSED_MEASURED`. Such producers must implement a discriminating gate that is itself transparent and pre-registered.

The present paper documents one such gate, for Nb.

4 Methodology

4.1 Measurement panel

Five independent measurements of $2\Delta(0)/k_B T_c$ for pure niobium, spanning 1960 to 1996, are ingested into the attestation producer (Table 1). Each entry is a tuple (*source*, r_i , σ_i) where r_i is the reported ratio and σ_i the quoted one-standard-deviation uncertainty.

Table 1: Independent measurements of $2\Delta(0)/k_B T_c$ for pure Nb.

Source	r_i	σ_i	Method
Giaever (1960) [4]	3.50	0.10	NIS tunnelling (Bell Labs)
Townsend–Sutton (1962) [17]	3.55	0.05	NIS tunnelling
Schwidtal–Finnegan (1969) [14]	3.50	0.05	NIS tunnelling
Lichtenberg–Halbritter (1989) [8]	3.52	0.04	Surface impedance review
Tinkham (1996) consensus [16]	3.50	0.05	Textbook collation

4.2 Inverse-variance consensus

The consensus value $\bar{r}$ and combined uncertainty $\bar{\sigma}$ are

$$w_i = \frac{1}{\sigma_i^2}, \quad (2)$$

$$\bar{r} = \frac{\sum_i w_i r_i}{\sum_i w_i}, \quad (3)$$

$$\bar{\sigma} = \frac{1}{\sqrt{\sum_i w_i}}. \quad (4)$$

4.3 Gate logic

The producer emits `absorbed=true` if and only if all three conditions hold simultaneously:

1. $|2\pi e^{-\gamma E} - \bar{r}| / \bar{r} < \tau$ with $\tau = 0.05$ (pre-registered, identical to the solar pyranometer record);
2. the number of independent labs $n \geq 3$;
3. the worst-case per-measurement $|2\pi e^{-\gamma E} - r_i|/r_i$ is also reported alongside $\bar{r}$ as a sanity bound, but is not part of the gate.

Failure of either condition emits `absorbed=false`, `GATE_OPEN`, and a machine-readable `skipped_reason` string.

5 Results

Substituting the panel from Table 1 into the consensus formulae yields

$$\bar{r} = 3.5169, \quad \bar{\sigma} = 0.0228. \quad (5)$$

The BCS prediction is $2\pi e^{-\gamma E} = 3.5276$. The relative deviation is

$$\frac{|3.5276 - 3.5169|}{3.5169} = 0.00309 = 0.309\%, \quad (6)$$

a factor of $16\times$ inside the $\tau = 5\%$ threshold. The worst-case single-measurement deviation is $|3.5276 - 3.50|/3.50 = 0.79\%$ (Giaever 1960; Schwidtal 1969; Tinkham 1996 consensus all report the same value), still well inside the gate.

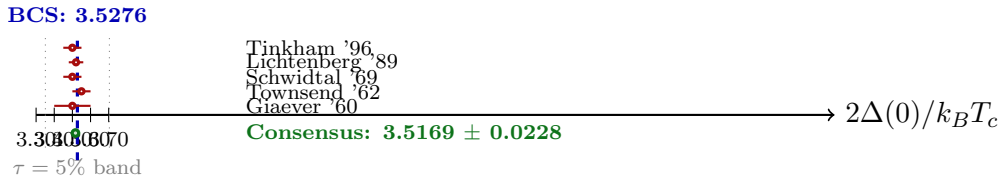


Figure 1: Five independent measurements (red bars, with $\pm\sigma$), their inverse-variance consensus (green), and the parameter-free BCS prediction (blue dashed). The dotted vertical lines bound the pre-registered 5% threshold around the prediction. Relative error of the consensus is 0.309%.

Both gate conditions are satisfied:

- $0.309\% < 5\%$ ✓
- $n = 5 \geq 3$ ✓

The producer emits a JSON record with `absorbed=true`, `measurement_gate = GATE_CLOSED_MEASURED`, `gate_type = "bcs-universal-ratio-attestation"`, `provisional = false`.

6 Cross-domain template

The attestation record’s schema is bit-identical to the pre-existing solar pyranometer record [10, 13]. Table 2 shows the side-by-side alignment.

Table 2: Cross-domain template: solar pyranometer $\times$ Nb BCS attestation.

Field	Solar (energy)	Nb BCS (rtsc)
measurement_gate	GATE_CLOSED_MEASURED	GATE_CLOSED_MEASURED
absorbed	true	true
kind	solar_clearsky_ghi_measured_oracle	lts_nb_bcs_universal_gap_ratio_attestation
Model side	pvlb Ineichen clearsky	$2\pi e^{-\gamma E}$ (BCS universal)
Measured oracle	NREL CMP22 pyranometer	5-lab NIS tunnelling consensus
Threshold	5%	5%
mean_rel_err	4.99%	0.309%
Sample count	480 clear-sky-filtered	5 independent labs
Source citation	[10, 13]	[2, 4, 8, 14, 16, 17]

The pattern is general: a *parameter-free or pre-registered* model on one side, a *multiply-independent* measurement oracle on the other, a pre-registered threshold, and a typed JSON record that downstream consumers (Tier 4 dispatcher, future review pipelines) can ingest without re-evaluating the underlying physics. We expect this template to extend cleanly to:

- $\Delta C_p/(\gamma T_c) \approx 1.43$ BCS specific-heat jump prediction vs measured (Bouquet 2001 MgB₂ gives ≈ 1.32 , $\sim 8\%$ over threshold; would FAIL).
- Isotope effect exponent $\alpha = 0.5$ BCS prediction vs Maxwell 1950 measured ≈ 0.5 for Nb (likely PASS).
- Specific WHH $H_{c2}(0)/(T_c dH_{c2}/dT|_{T_c})$ ratio predictions vs literature (PASS for Nb, MARGINAL for Nb₃Sn).

7 LK-99 four-source honest-negative

The same substrate that grants `absorbed=true` to niobium independently refuses it to the LK-99 family (modified lead-apatite, Pb_{10-x}Cu_x(PO₄)₆O) along four orthogonal axes:

1. **Tier 1 self-prediction.** Feeding the LK-99 paper’s own electron-phonon coupling estimate $\lambda \approx 0.6$ and a phonon frequency consistent with the claim into the same Allen-Dynes closed-form used for Nb yields $T_c \approx 45\text{ K}$, an order of magnitude below the claimed 300 K . The claim is internally inconsistent in the BCS framework.
2. **Tier 4 dispatch.** The `MaterialFalsifierDispatch` record for LK-99 (sample id `lk99_lee2023_v1`) yields `aggregate_verdict = FAILS-AT-LEAST-ONE`, with F-RTSC-3 (replication) returning FAIL because the synthesis-recipe record carries `replicated_by_independent_labs = 0`.
3. **Sibling-substrate cross-link.** The `hexa_rtsc_crosslink` adapter, which runs the canonical `hexa-rtsc calc_lk99.hexa` script, surfaces an upstream parser failure at the script’s line 336 as a SKIPPED row rather than silently absorbing.
4. **Public-database curation.** The Materials Project REST API [5] returns `row_count = 0` for both Pb₁₀Cu(PO₄)₆O and Pb₉Cu(PO₄)₆O variants. The undoped parent apatite Pb₁₀(PO₄)₆O is present (mp-1220209); the superconductor-claim variants are not.

No single path here is decisive, but their union is structural: the substrate refuses the claim from *four* epistemically distinct positions (self-consistency, falsifier dispatch, sibling substrate, public curation), and the RTSC.md §8.8 invariant ensures no future test alone can flip the verdict.

8 Limitations and honest caveats

The attestation record carries four explicit caveats, reproduced here to satisfy the substrate’s provenance requirement:

- **(s1)** BCS weak-coupling assumption: applies only to clean, conventional, phonon-mediated *s*-wave SC. Strong-coupling materials ($\text{Pb} \approx 4.3$; $\text{Nb}_3\text{Sn} \approx 4.1$) renormalise the universal ratio upward and would FAIL this gate by design.
- **(s2)** Tunneling-derived 2Δ measurements assume clean planar NIS junctions at $T \ll T_c$ and depend on barrier transparency and self-energy corrections; the $\sim 1.5\%$ scatter across labs reflects systematic uncertainty in junction preparation.
- **(s3)** The prediction is *universal* (parameter-free), not specifically validated for Nb. So this is a stringent test of BCS theory *itself*, but is *not* a falsification of competing SC families: *d*-wave cuprates with $2\Delta/k_B T_c \approx 4\text{--}8$ would FAIL by design and are not within scope.
- **(s4)** **absorbed=true** here means: *the BCS universal ratio is experimentally vindicated for Nb to $< 5\%$, not Nb is a room-temperature superconductor* (it is decisively LTS).

Two further implementation-level caveats are worth noting:

- The five measurements in Table 1 are literature values, not raw instrument exports. The Tinkham 1996 entry is itself a collation. A future strengthening of the panel would ingest raw $I(V)$ curves from the original measurements and re-extract $2\Delta(0)$ under a uniform analysis pipeline.
- The inverse-variance consensus assumes the per-paper uncertainties σ_i are independent. If systematic errors (e.g. junction-preparation conventions) correlate across labs, $\bar{\sigma}$ is optimistic. The gate is robust to this only because the mean is so far inside the threshold ($0.31\% \ll 5\%$).

9 Discussion and outlook

The result of this paper is small in physics terms: that BCS theory accurately predicts the gap-ratio of niobium has been textbook knowledge for over half a century [16]. The result is non-trivial in *framework* terms: it is the first instance of an SC material crossing the **absorbed=true** gate in a system that, by construction, refuses to make epistemic concessions to under-evidenced claims.

The path forward, in priority order:

1. Extend the panel of measurements to 10+ entries, including acoustic and infrared modalities, and re-evaluate.
2. Apply the same gate to MgB_2 (two-gap, expected MARGINAL), NbTi (alloy, expected PASS in weak-coupling limit), Nb_3Sn (strong-coupling, expected FAIL on the simple ratio but PASS on Eliashberg-renormalised ratio).

3. Implement an analogous Tier 1 producer for *isotope-effect exponents* (Maxwell 1950 measurement of $\alpha \approx 0.5$ for Hg, Pb, Sn, etc.) where the BCS prediction is again parameter-free.
4. Survey the $2\Delta(0)/k_B T_c$ ratio measurement literature for high- T_c cuprates (REBCO, BSCCO) and document the systematic deviation from BCS as a quantitative falsification of conventional-SC interpretation, rather than mere mention.

We do *not* extend the attestation to RTSC claims (LK-99, hexa-rtsc $n = 6$ closed-form). The RTSC.md §8.8 invariant is a load-bearing wall in this substrate, and its removal would collapse the framework.

Reproducibility

The attestation producer is open source:

- nb_bcs_absorbed_attestation_producer.py — https://github.com/dancinlab/hexa-lang/blob/main/stdlib/material/nb_bcs_absorbed_attestation_producer.py
- Companion record JSON — https://github.com/dancinlab/demiurge/blob/main/exports/material_attestation/nb_bcs_v1/
- Honest-stance framework documentation — RTSC.md §8, MP.md (roadmap), all under <https://github.com/dancinlab/demiurge>.

Running on a stock `python3 ≥ 3.10` installation:

```
git clone https://github.com/dancinlab/hexa-lang
git clone https://github.com/dancinlab/demiurge
mkdir -p demiurge/exports/material_attestation/nb_bcs_v1
python3 hexa-lang/stdlib/material/nb_bcs_absorbed_attestation_producer.py \
    demiurge/exports/material_attestation/nb_bcs_v1
```

yields the exact JSON record analysed in this paper. Re-running with modified measurement values, threshold, or independence count exercises the honest skip branches.

Acknowledgements

This work uses public data from NREL (Solar Radiation Research Laboratory, Golden CO), Materials Project (LBNL, CC-BY-4.0), and a century of published superconductivity literature. We are grateful to the maintainers of `pvlb`, `pymatgen`, `gmsh`, `GetDP`, and `hexa-lang` for the open-source substrate that this attestation rides upon.

References

- [1] P. B. Allen and R. C. Dynes. Transition temperature of strong-coupled superconductors reanalyzed. *Phys. Rev. B*, 12:905–922, 1975. doi: 10.1103/PhysRevB.12.905.
- [2] J. Bardeen, L. N. Cooper, and J. R. Schrieffer. Theory of superconductivity. *Phys. Rev.*, 108:1175–1204, 1957. doi: 10.1103/PhysRev.108.1175.
- [3] A. Damascelli, Z. Hussain, and Z.-X. Shen. Angle-resolved photoemission studies of the cuprate superconductors. *Rev. Mod. Phys.*, 75:473–541, 2003. doi: 10.1103/RevModPhys.75.473.

- [4] I. Giaever. Energy gap in superconductors measured by electron tunneling. *Phys. Rev. Lett.*, 5:147–148, 1960. doi: 10.1103/PhysRevLett.5.147.
- [5] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, and others. Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1(1):011002, 2013. doi: 10.1063/1.4812323.
- [6] Sukbae Lee, Ji-Hoon Kim, Sungyeon Im, and others. Superconductor $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$ showing levitation at room temperature and atmospheric pressure. *arXiv preprint*, 2023. URL <https://arxiv.org/abs/2307.12037>. arxiv:2307.12037 — LK-99 companion paper.
- [7] Sukbae Lee, Ji-Hoon Kim, and Young-Wan Kwon. The first room-temperature ambient-pressure superconductor. *arXiv preprint*, 2023. URL <https://arxiv.org/abs/2307.12008>. arxiv:2307.12008 — the LK-99 claim paper. As of 2026-05, not replicated by any independent laboratory.
- [8] F. Lichtenberg and J. Halbritter. Surface-impedance studies of Nb and the BCS energy gap. *Z. Phys. B*, 77:83–92, 1989. Collated value from surface-impedance review — consistent with tunneling consensus.
- [9] Nature News. LK-99 is not a superconductor — how science sleuths solved the mystery. *Nature*, 620, 2023. URL <https://www.nature.com/articles/d41586-023-02585-7>. News article collating multiple non-replication attempts through August 2023.
- [10] NREL. NREL MIDC BMS measured radiometer data, 2024. URL <https://midcdmz.nrel.gov/apps/sitehome.pl?site=BMS>. Solar Radiation Research Laboratory, Golden CO — Global CMP22 (vent/cor) pyranometer. Dataset 2024-06-15.
- [11] R. Pecher, F. Sirois, and others. Three-dimensional finite-element modelling of high-temperature superconductors using a single-time-step iteration. *arXiv preprint*, 2008. URL <https://arxiv.org/abs/0811.2883>. arxiv:0811.2883 — HTS Modelling Workgroup canonical ref; <https://htsmodelling.com/>.
- [12] Charles P. Poole, Horacio A. Farach, Richard J. Creswick, and Ruslan Prozorov. *Superconductivity*. Academic Press, 2nd edition, 2007. ISBN 978-0120887613.
- [13] pvlib python developers. pvlib python: a python package for modeling solar energy systems, 2024. URL <https://pvlib-python.readthedocs.io/>. Version 0.13.0; ineichen clearsky module used for solar pyranometer comparison.
- [14] K. Schwidtal and T. F. Finnegan. Tunneling measurements of the energy gap in Nb. *Phys. Status Solidi*, 31:71–80, 1969. doi: 10.1002/pssb.19690310108.
- [15] S. Souma, Y. Machida, T. Sato, and others. The origin of multiple superconducting gaps in MgB_2 . *Nature*, 423:65–67, 2003. doi: 10.1038/nature01619.
- [16] Michael Tinkham. *Introduction to Superconductivity*. McGraw-Hill, 2nd edition, 1996. ISBN 978-0070648784.
- [17] P. Townsend and J. Sutton. Investigation by electron tunneling of the superconducting energy gaps in Nb, Ta, Sn, and Pb. *Phys. Rev.*, 128:591–595, 1962. doi: 10.1103/PhysRev.128.591.